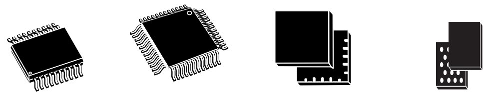
STM32C051x6/x8

Arm®Cortex®-M0+ 32-bit MCU, 64 KB flash, 12 KB RAM, 2 x USART, timers, ADC, comm. I/Fs, 2-3.6V

Datasheet - production data

Features

- Includes ST state-of-the-art patented technology
- Core: Arm® 32-bit Cortex®-M0+ CPU, frequency up to 48 MHz
- -40°C to 85°C/105°C/125°C operating temperature
- Memories
  - Up to 64 Kbytes of flash memory with protection and securable area
  - 12 Kbytes of SRAM with hardware parity check
- CRC calculation unit
- Reset and power management
  - Voltage range: 2.0 V to 3.6 V
  - Power-on / power-down reset (POR/PDR)
  - Programmable brownout reset (BOR)
  - Low-power modes:
Sleep, Stop, Standby, Shutdown
- Clock management
  - 4 to 48 MHz crystal oscillator
  - 32 kHz crystal oscillator with calibration
  - Internal 48 MHz RC oscillator (±1 %)
  - Internal 32 kHz RC oscillator (±5 %)
- Up to 45 fast I/Os
  - All mappable on external interrupt vectors
  - All 5 V-tolerant
- 5-channel DMA controller with flexible mapping
- 12-bit, 0.4 µs ADC (up to 19 ext. channels)
  - Conversion range: 0 to 3.6 V
- 9 timers: 16-bit for advanced motor control, one 32-bit timer and four 16-bit general-purpose, two watchdogs, SysTick timer
- Calendar RTC with alarm



TSSOP20 (6.5 × 4.4 mm)

LQFP32 (7 × 7 mm)

LQFP48 (7 × 7 mm)

UFQFPN28 (4 × 4 mm)

UFQFPN32 (5 × 5 mm)

UFQFPN48 (7 × 7 mm)

WLCSP15 (2.25 x 1.40 mm)

- Communication interfaces
  - Two I²C-bus interface supporting Fast-mode Plus (1 Mbit/s) with extra current sink; one supporting SMBus/PMBus™ and wake-up from Stop mode
  - Two USARTs with master/slave synchronous SPI; one supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wake-up feature
  - Two SPIs (24 Mbit/s) with 4- to 16-bit programmable bitframe, one multiplexed with I²S interface; two extra SPIs through USARTs
- Development support: serial wire debug (SWD)
- 96-bit unique ID
- All packages ECOPACK 2 compliant

Table 1. Device summary

Reference	Part number
STM32C051x6	STM32C051C6, STM32C051F6, STM32C051G6, STM32C051K6
STM32C051x8	STM32C051C8, STM32C051F8, STM32C051G8, STM32C051K8, STM32C051D8